THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 81	First term + (12-1) x 7 = 4+11x7
2	**(a)** 240	The two angles between the hands add up to 360°.
3	**(d)** 1234	End with the smallest even digit, then arrange the rest in increasing order.
4	**(c)** 100	Add the two biggest numbers.
5	**(c)** 10	$48 = 2^4 \times 3$; use $(e_1+1)(e_2+1)\ldots$
6	**(c)** 18	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
7	**(d)** 120	6 degrees per minute, or 30° per number on the dial.
8	**(c)** 90	First digit 1-9 and middle digit 0-9, the last digit is fixed: 9×10.
9	**(c)** 75	Find the difference in symbols then multiply.
10	**(b)** 12	Highest common factor.
11	**(d)** 156	Sum of first n even numbers is $n \times (n+1)$.
12	**(b)** 5	30° per hour.
13	**(d)** 29992	Fill the middle digits with 9.
14	**(b)** 165	Total height times scale.
15	**(d)** 36	Lowest common multiple.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Rule: 20th term = first term $+(20-1)\times$ difference $= 5 + 19 \times 3$ (1); 20th term $= 62$ (1).	1+1
17	Smaller angle $= 150^\circ$ (obtuse) (1); Larger angle $= 360 - 150 = 210^\circ$ (reflex) (1).	1+1
18	Greatest 97431 (1); Smallest 13479 (1).	1+1
19	Highest 75, lowest 35; Difference $= 40$ (1); Total $= 180$ (1).	1+1
20	$24 = 2\times2\times2\times3$, $36 = 2\times2\times3\times3$ (1); HCF $= 12$, LCM $= 72$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sticks for 12 triangles $= 3 + 2 \times (12-1) = 25$ (1); For 35 sticks: $3 + 2 \times (m-1) = 35$, so $2 \times (m-1) = 32$ (1); $m - 1 = 16$, so 17 triangles (1).	1+1+1
22	Three-quarter turn $= 270^\circ$ and quarter turn $= 90^\circ$ (1); Net turn $= 270 - 90 = 180^\circ$ (1); Type: straight (1).	1+1+1
23	Step 1: 8172 (1); Step 2: 7443 (1); Digits used: arrange 8172 to get the next number (1).	1+1+1
24	Total $= 130$ (1); Fraction $= \frac{25}{130} = \frac{5}{26}$ (1); Others $= 105$ (1).	1+1+1
	OR: $180 = 2^2 \times 3^2 \times 5$ (1); Number of factors $= (2+1)\times(2+1)\times(1+1) = 18$ (2).	1+2
25	There are 30 odd numbers, sum $= 900$ (1); Even numbers: sum $= 30 \times 31$ (1); $= 930$ (1).	1+1+1
	OR: Anticlockwise total $= 60 + 70 = 130^\circ$ (1); Clockwise 40° is subtracted (1); Net turn $= 90^\circ$ anticlockwise (right) (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	10th figure $= 5 + (10-1) \times 4 = 41$ (1); More than 91 needs $(91-5) \div 4$ more steps (1); Steps needed: 13 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
	OR: Hours passed $= 2$, hour hand turns 60° (1); Minute hand turns 2 full turns $= 720^\circ$ (1); At 3:00 the hour hand points to 3 ($= 90^\circ$) (1); Smaller angle between the hands $= 90^\circ$ (1); Type: right (1).	1+1+1+1+1

27	The last digit is also 2, and the middle two digits are equal: $2 \times 2 + 2m = 10$ so $m = 3$ (1); Number 2332 (1); Reverse is the same: palindrome (1); Add 11: 2343 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Mon $= 36$, Tue $= 72$, Wed $= 108$ (1); Thursday $= 36 + 72 = 108$ (1); Total for four days $= 36 + 72 + 108 + 108$ (1); $= 324$ (1); Symbols needed for Thursday: 9 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 210	Add 20.
2	**(c)** 160	Full turn is 360°.
3	**(d)** 4	Count each step.
4	**(c)** $\frac{6}{29}$	Fraction $= \frac{30}{145}$ (simplified)
5	**(c)** 1	Digit sum must be a multiple of 3.
6	**(b)** 21	Sticks $= 3 + 2 \times (n-1) = 2n+1$.
7	**(a)** 27	A bisector divides the angle into two equal parts.
8	**(c)** 1441	Keep the first half and mirror it; if that is not bigger, increase the middle.
9	**(a)** 65	Add the symbols, then multiply by the value of one symbol.
10	**(a)** 994	Divide 999 by the number.
11	**(d)** 625	Consecutive triangular numbers add to a square.
12	**(b)** 270	Add the two turns in degrees.
13	**(c)** 465	Pair the numbers: first with last.
14	**(a)** 0	Count.
15	**(d)** 9	Count them carefully.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Terms are 3, 7, ... each 4 more than the last (1); 35 is a term but 36 lies between two terms, so it is not a term (1).	1+1
17	Bisector gives two equal angles (1); $\angle AOC = 2 \times 28 = 56^\circ$, type acute (1).	1+1
18	87, 165, 726, 1353, 4884 (1); Palindrome 4884 (1).	1+1
19	VI-A $= 50$ students; VI-B $= 70$ students (1); Symbols $= 7$ (1).	1+1
20	1234 divided by 9 leaves remainder 1 (1); Add 8 to get 1242 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Largest below: 1221 (1); Smallest above: 1331 (1); Check by reversing (1).	1+1+1
22	Total $= 280$ kg (1); Heights: 2, 3, 4, 5 cm (1); Tallest bar: 5 cm (week 4) (1).	1+1+1
23	LCM of 12 and 18 is 36 (1); After 36 minutes (1); Time: 9:36 (1).	1+1+1
24	$T_{20} = 210$, $T_{21} = 231$ (1); Sum $= 441$ (1); $21^2 = 441$ matches (1).	1+1+1
	OR: Total segments $= \frac{5\times4}{2} = 10$ (1); Segments having A as an end point: 4 (1); Segments without A: $10 - 4 = 6$ (1).	1+1+1
25	Pairs sum to 41 (1); Number of pairs $= 40 \div 2 = 20$ (1); Sum $= 20 \times 41 = 820$ (1).	1+1+1
	OR: Count more than 2: 8 (1); Count exactly 2: 2 (1); Use tally marks to count carefully (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Side $= \mathrm{HCF}(450, 360) = 90$ cm (1); Along the length: 5 tiles (1); Along the width: 4 tiles (1); Total $= 20$ tiles (2).	1+1+1+2
	OR: Sum of first 9 numbers $= \frac{9\times10}{2} = 45$ (1); Sum of cubes $= 2025$ (1); $45^2 = 2025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
27	$\angle AOD = 30 + 45 + 60 = 135^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 135 = 45^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
	OR: Try arrangements (1); Closest number is 5137 (2); Difference $= 137$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** |
Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 128	Each term is double the one before it.
2	**(d)** right	Find the angle first.
3	**(c)** 6174	Largest - smallest.
4	**(c)** 10	Convert the increase into cm.
5	**(c)** 17	$210 = 2 \times 3 \times 5 \times 7$
6	**(a)** 93	First term + (15-1) x 6 = 9+14x6
7	**(a)** 75	Straight angle is 180°.
8	**(b)** 9100	Put 9s from the left until the sum is used up.
9	**(b)** 135	Add the two biggest numbers.
10	**(c)** 8	$54 = 2 \times 3^3$; use $(e_1+1)(e_2+1)\ldots$
11	**(b)** 25	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
12	**(c)** 210	The two angles between the hands add up to 360°.
13	**(b)** 2576	End with the smallest even digit, then arrange the rest in increasing order.
14	**(a)** 125	Find the difference in symbols then multiply.
15	**(b)** 15	Highest common factor.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Odd: $30^2 = 900$ (1); Even: $30 \times 31 = 930$ (1).	1+1
17	Hour hand: $4 \times 30 = 120^\circ$ (1); Minute hand: 4 full turns $= 1440^\circ$ (1).	1+1
18	Second row: 5, 4, 5; Third row: 9, 9 (1); Top $= 18$ (1).	1+1
19	Count each value; 1: 2; 2: 2; 3: 2; 5: 3; 6: 5 (1); Most repeated: 6 (1).	1+1
20	Digit sum of known digits is 15 (1); Missing digit $= 3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Length of piece $= \text{HCF}(48, 72) = 24$ m (1); Pieces from the first rope: 2; from the second: 3 (1); Total pieces $= 5$ (1).	1+1+1
22	Sticks for 15 triangles $= 3 + 2 \times (15 - 1) = 31$ (1); For 41 sticks: $3 + 2 \times (m - 1) = 41$, so $2 \times (m - 1) = 38$ (1); $m - 1 = 19$, so 20 triangles (1).	1+1+1
23	Full turn $= 360^\circ$ and quarter turn $= 90^\circ$ (1); Net turn $= 360 - 90 = 270^\circ$ (1); Type: reflex (1).	1+1+1
24	Step 1: 6264 (1); Step 2: 4176 (1); Digits used: arrange 6264 to get the next number (1).	1+1+1
	OR: Total $= 95$ (1); Fraction $= \frac{50}{95} = \frac{10}{19}$ (1); Others $= 45$ (1).	1+1+1
25	$200 = 2^3 \times 5^2$ (1); Number of factors $= (3+1) \times (2+1) = 12$ (2).	1+2
	OR: There are 40 odd numbers, sum $= 1600$ (1); Even numbers: sum $= 40 \times 41$ (1); $= 1640$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Total $= 115$ (1); Maximum: Kabaddi with 60 (1); Fraction $= \frac{60}{115} = \frac{12}{23}$ (1); Scale 1 cm = 10 students: height $= 6$ cm (2).	1+1+1+2
	OR: LCM of 4, 5, 6 $= 60$ (1); Smallest number $= 61$ (1); Next number $= 121$ (1); Check remainders (2).	1+1+1+2
27	12th figure $= 3 + (12 - 1) \times 5 = 58$ (1); More than 108 needs $(108 - 3) \div 5$ more steps (1); Steps needed: 11 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1

	OR: Hours passed $= 5$, hour hand turns 150° (1); Minute hand turns 5 full turns $= 1800^\circ$ (1); At 7:00 the hour hand points to 7 ($= 210^\circ$) (1); Smaller angle between the hands $= 150^\circ$ (1); Type: obtuse (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND